

Reduction Series Master Ledger

Auditability and Version-Control Pass

(Result Ledger, Run Provenance, Raw P_{solve} , Notation, Capacity-Law Convention, Dependency Graph)

Tempotron learning-surface program

2026-06-28

Abstract

This document is the single auditability reference for the twelve-paper tempotron-capacity reduction series (`academic/NN-*.tex`, $NN = 01..12$), produced in response to a reviewer request for a version-control / auditability pass. It collects, in one place: (1) a **result ledger** with the authoritative status and run configuration of each reduction; (2) a **run-provenance table** of the confounded/buggy runs and their fixes; (3) the **raw $P_{\text{solve}}(\alpha)$ tables** that exist as committed sweep outputs, with file paths, and an honest list of the crossings for which no raw committed table was located; (4) a **notation table**; (5) a **capacity-law master convention** standardising the vocabulary the reviewer flagged (EVT lift, “total capacity”, and the status of the +2.58 constant); and (6) a **dependency graph** of the reductions. Field values are pulled from each reduction’s `.tex` (abstract, methods/reproducibility) and the `derivations/` notes; fields that are genuinely not recorded are marked “n/r” (not recorded) and missing raw tables are flagged rather than invented.

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Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in `lit/tempotron-reductions/derivations/`.

1 Result ledger

The ledger is split into two keyed tables that share the reduction number: Table 1 (identity, authoritative status, superseded runs, final crossing) and Table 2 (run configuration). “Final crossing” is the authoritative findable $\hat{\alpha}_c$ (or, for #2/#3, the existence/geometry observable that is the paper’s result). Values are quoted from the cited paper’s abstract and experiment/methods sections.

Notes on the ledger.

- #2 and #3 do not report a learned $\hat{\alpha}_c$ crossing: #2 measures a *convex existence* capacity by max-margin QP (no learner, no findable/existence gap), and #3 measures the shattering order parameter $q_0(K)$ on the #1 runs (its capacity statement is the analytic $\alpha_c = (\ln \ln K)/(2 \ln 2)$ lift, shared with #1).
- #9 and #12 report a closed-form *replica-symmetric* radial value $\alpha_c^{\text{RS}} = 4.598$ (4.660 threshold-optimised) in addition to the findable lab crossing; this is a derivation, not a learned crossing (§2, §5).
- Density convention differs across campaigns: #4/#6/#7 hold 2.0 spikes/afferent; #8–#12 (per-kernel campaign) hold 3.0. All are *fixed* across the swept axis (the density-confound fix).

2 Run-provenance table (confounded / fixed runs)

Table 3 records the runs and framings that were wrong and their fixes, sourced from the affected papers’ Errata / Experimental / Limitations sections (notably 07-kernel-universality.tex §Errata).

3 Raw $P_{\text{solve}}(\alpha)$ tables

The reviewer asked for the raw solved-fraction-vs- α data behind each crossing, not just the interpolated $\hat{\alpha}_c$. **Raw committed sweep outputs now exist for all reductions #1–#12** (plus the `tempotron_capacity` program). #1–#6 were committed with their original campaigns; the per-kernel / cross-kernel campaigns (#7–#12), whose committed artifacts were previously only summary $\hat{\alpha}_c$ strings, were recovered in a 2026-06-28 **Tier-1 reproducibility pass** (the per-seed `results.csv` files, retained in the lab run store but never committed, were extracted into the repo — *no re-run*). Two cells whose original runs `timed_out` remain unbacked and are named honestly below.

3.1 Raw tables that were located (with provenance paths)

All paths are relative to the repository root.

- **#1 discrete:** per-seed rows (`seed,K,rho,beta,N,alpha,P,solved,...`), aggregate to $P_{\text{solve}}(\alpha) = n_{\text{solved}}/n_{\text{seeds}}$ by grouping on α : `experiments/discrete_capacity/results/m1_K{2,4,16,32,64}.csv`; `finite-N`: `m1_finiteN_N250_K4-16.csv`, `m1_finiteN_N250_K64.csv`, `m1_finiteN_N1000_K4.csv`, `m1_finiteN_N1000_K16.csv`. The paper quotes the aggregated descents verbatim, e.g. $K=2$: $P_{\text{solve}} = 1.0, 1.0, 1.0, 0.125$ at $\alpha = 2.0, 2.2, 2.4, 2.6$; $K=16$: $1.0, 1.0, 0.833, 0.0$ at $\alpha = 3.1, 3.4, 3.7, 4.0$.

- **#2 spherical:** separability indicators (`seed,N,kappa,alpha,P,sep_unc,sep_pos,...,p_silent`), separable-fraction= P_{solve} : `experiments/spherical_capacity/results/m2_signconstrained_N400.csv`, `m2_finiteN_ladder.csv` ($\alpha \in [0.06, 2.2]$, 29 values).
- **#3 EVT:** same schema as #1 (`...,alpha,P,solved,...`): `experiments/evt_capacity/results/m3_q0_ladder` controls `m3_q0_rho0.csv`, `m3_q0_rho1ctrl.csv`, finite- N `m3_finiteN_N{250,1000}*.csv`. (#3's headline is q_0 , but these carry the per-seed solve data.)
- **#4 GP:** `experiments/gp_capacity/results/m4_cap_taus{1.5,4.5,15}.csv` (`...,alpha,P,solved,...`); the paper's Methods reproduces the grids, e.g. $\tau_s=15$: $P_{\text{solve}} = 1.00, 0.94, 0.69, 0.19, 0.00, 0.00$ at $\alpha = 1.8\text{--}2.8$. Witness (not P_{solve}): `m4_witness.csv`.
- **#5 soft-max:** `experiments/softmax_capacity/results/K2_r0.csv`, `K3_r0.csv`, `K3_r0.3.csv`, `r0_b{0.5,1.0,1.5}.csv`, `r0.3_b1.0.csv`; finite- N `m5_finiteN_b0.5_N{250,500,750}.csv` (`...,alpha,P,solved,...`).
- **#6 sinusoidal:** `experiments/sinusoidal_capacity/results/corrected/K{8,16,32,64}_d*.csv`; ladder/ (12 timestamped CSVs); finite- N `m6_finiteN_N{500,1000,1500}*.csv` (`seed,Keff_target,Keff`).
- **tempotron_capacity (already-aggregated):** explicit `n_solved,p_solve` columns: `experiments/tempotron` `t2c_n1000/results.csv`, `t2_faithful_fig4a_n500/fig4a.csv` (`alpha,n_solved,n_seeds,...`), and siblings (`t2c_smalln_*`, `t2b_synchrony`, `v12_j*`). Relevant as the faithful-tempotron provenance underlying #1/#3.
- **#7–#12 (Tier-1 pass, 2026-06-28):** per-seed/per- α rows (`seed,kernel,Keff_target,Keff_measured,I`) under `experiments/kernel_universality_capacity/results/`: subdirs `v13_xkernel_density2/` (#7 cross-kernel, $N=500$, density 2.0; gauss/sinc/alpha/gabor/sinusoid $\times K_{\text{eff}} \in \{8, 16, 32, 64\}$, gabor split per $Q \in \{2, 4, 8\}$) and `v14_perkernel_density3/` (#8 gauss / #10 alpha / #11 rect / #12 sinusoid + #9 gabor preflight, density 3.0). **50 cell CSVs, 4501 rows**; index in `results/MANIFEST.csv` (cell $\rightarrow$ source lab job IDs) and `results/README.md`. Assembled + re-verified by `experiments/kernel_universality_capacity/assemble_results.py` (`--verify` re-derives each $\hat{\alpha}_c$; all match Table 1, e.g. #8 gauss 2.46/2.68/2.91/3.16, #7 gabor-Q4 3.49/3.64/3.89).

3.2 Residual gaps after the Tier-1 pass (named honestly)

The previous outstanding reproducibility item — regenerating and committing per-seed P_{solve} CSVs for #7–#12 under `experiments/kernel_universality_capacity/results/` — is **now closed** (§3.1). Only `status=succeeded` runs were committed (a `timed_out` run has incomplete seeds/ α and would bias P_{solve}), so **two published cells whose original runs timed out remain unbacked**:

- **#9 Gabor matched- K_{eff} ($N=500$, density 3.0):** every cell `timed_out` — and these are the same cells the `omega0=argv` bug railed to $K_{\text{eff}} \approx 499$ (§2). So the matched- K_{eff} band-pass > low-pass isolation has no clean committed data; as Table 1 already states, #9's matched- K_{eff} result *rests on* #7 (committed here, per- Q). What *is* committed for #9: the $N=200$ narrowband preflight (findable ≈ 3.2). Re-running these cells with the fixed grid + longer timeout is named future work.
- **#12 pure sinusoid $K_{\text{eff}}=8$ (density 3.0):** the sole $N=500$ $K_{\text{eff}}=8$ run `timed_out`; the published 3.40 for that one cell is unbacked. The flatness claim still rests on the committed $K_{\text{eff}} \in \{16, 32, 64\}$ (≈ 3.4).

A handful of committed #7 cells carry 8 seeds (not 16) where one shard timed out, and gabor $Q2, K_{\text{eff}}=32$ never crosses in-grid (blank `half_crossing_ac` in `MANIFEST.csv`) — both already noted in the #7 paper’s limitations.

4 Notation table

5 Capacity-law master convention

Standardised vocabulary (reviewer-flagged). Use these terms uniformly across the series.

1. **EVT lift.** The asymptotic capacity lift above the perceptron anchor is the *double* logarithm

$$\Delta\alpha_c^\infty = \frac{\ln \ln K}{2 \ln 2}, \quad \text{slope } \frac{d\alpha_c}{d \ln \ln K} = \frac{1}{2 \ln 2} = 0.721.$$

This is the genuine asymptotic object (the log of the solution-cluster count, #3). An apparent single logarithm $\ln K$ over the accessible range ($K_{\text{eff}} \lesssim 32$) is the *predicted pre-asymptotic / finite- K model-selection artifact* (#8, note 13): $\ln \ln K$ spans only 0.33–1.24 there, so a straight line in $\ln K$ absorbs the gentle rise.

2. **“Total capacity” (anchored form).** Call α_c a “total capacity” *only* when the intercept is specified. The canonical anchored form is #8’s

$$\alpha_c(K_{\text{eff}}) = 2 + A \ln \ln K_{\text{eff}}, \quad A \rightarrow \frac{1}{2 \ln 2} \text{ (asymptotic),}$$

with the *perceptron 2 as a fixed (unfitted) anchor* (Cover/Gardner). A fit with a free intercept is a “lift fit”, not a total capacity.

3. **The +2.58 constant is NOT asymptotic.** The additive $\alpha_0 \approx 2.58$ that appears in $\alpha_c(K) = (\ln \ln K)/(2 \ln 2) + 2.58$ is a *small- K numerical fit* to the Rubin et al. replica capacity at $K = 2, 3, 4$. It is *not* a theoretical or asymptotic constant and does *not* converge as $K \rightarrow \infty$: the true next term is a third iterated logarithm, $\alpha_c(K) = \frac{\ln \ln K}{2 \ln 2} + \frac{\ln \ln \ln K}{\ln 2} - 0.383 + o(1)$ (codex cross-check, #1/#3). Cite 2.58 only as a finite- K benchmark offset (as in #11), never as a constant.
4. **Radial (two-quadrature) capacity.** The band-pass/sinusoid radial classifier $\sqrt{(\mathbf{w} \cdot \mathbf{a})^2 + (\mathbf{w} \cdot \mathbf{b})^2} > \vartheta$ has RS capacity $1/\alpha_c = \frac{1}{2}(\vartheta^2 - \sqrt{2\pi}\vartheta + 2)$, i.e. $\alpha_c^{\text{RS}} = 4.598$ at the balanced threshold (4.660 threshold-optimised). The older $\alpha_c = 4$ is the *one-quadrature* squared-perceptron value (a different geometry) and is superseded for this readout (#9/#12).

6 Dependency graph

The series preamble does not load `tikz`, so the dependency structure is given as an explicit “X depends on Y” list and a compact adjacency table.

- **Spine.** #1 (discrete max-of- K) $\rightarrow$ #3 (EVT max-of-fields) $\rightarrow$ {#4 (GP/frozen-kernel), #6 (band-limited sinusoidal), #8 (Gauss), #11 (rect)}. The EVT lift derived in #3 (and its $K=1$ anchor from #1) is the engine every kernel reduction plugs its own K_{eff} into.

- **Cross-kernel hub.** #7 (cross-kernel universality) *uses* {#6, #8, #10, #11, #12} as its per-kernel inputs (matched- K_{eff} comparison across shapes).
- **Radial supersession.** #9 (Gabor) and #12 (pure sinusoid) *supersede* the earlier radial-bound language in #7: the one-quadrature “ $\alpha_c = 4$ ” is replaced by the two-quadrature $\alpha_c^{\text{RS}} = 4.598$ derived in #9/#12.
- **Saturation lineage.** #11 (rect) is the sharp ($\tau_s = 0$) limit of #4’s participation-ratio saturation; both contrast $K_{\text{eff}}^{\text{PR}}$ (finite) against the divergent Rice count.
- **Standalone anchors.** #2 (sign-constrained spherical perceptron) and #5 (soft-max β -family) are comparatively standalone: #2 is the rigid convex null control (halving of Gardner), #5 the connected-to-shattered interpolation that contains the #1/#3 hard-max as its $\beta \rightarrow \infty$ endpoint and the perceptron as its $\beta \rightarrow 0$ endpoint.

Sources. Every field is pulled from the cited reduction’s `academic/NN-*.tex` (abstract, Experimental Test, Methods/Reproducibility, Errata) and the `lit/tempotron-reductions/derivations/NN-*.md` prediction notes; raw-data paths are committed under `experiments/*_capacity/`. Fields marked “n/r” are not recorded in the source paper; raw tables marked “not located” were absent from committed outputs as of 2026-06-28.

Table 1: Result ledger A — identity, status, superseded runs, authoritative crossing. Status uses the paper’s own **Status:** tag.

#	kernel model	/	status	superseded runs	authoritative final crossing $\hat{\alpha}_c$
01	discrete (binned) max- of- K		PARTIAL	early ladders (patience<conv. time, low α -grid); $K=8$ dropped	2.514/2.945/3.820/4.050/4.150 at $K=2/4/16/32/64$ ($N=500$); fit $2.79 + 0.97 \ln \ln K$
02	sign- constrained spherical per- ceptron		CONFIRMED (STRONG)	refuted κ -dependent p_{act} conjecture	ratio $\alpha_c^+/\alpha_c^{\text{unc}} = 0.497 \pm 0.021$ ($= \frac{1}{2}$); $\alpha_c^+(0) = 1.003$; $p_{\text{silent}} = 0.501$
03	EVT max-of- K Gaussian fields		PARTIAL	retracted α_c values 1.20/1.99/2.58/3.11 (V2); reuses #1 runs	shattering $q_0(K)=1.00 \rightarrow 0.14$ ($K=1 \rightarrow 64$, $N=500$); $\alpha_c =$ $(\ln \ln K)/(2 \ln 2)$
04	GP / frozen- kernel (double-exp)		PARTIAL	—	$\hat{\alpha}_c = 2.275/2.360/2.345$ at $\tau_s=15/4.5/1.5$ ms ($N=400$); witness $\mathbb{E}[\max V]$ plateau
05	soft-max / surrogate (β -family)		PARTIAL (SOLID end- points + small-β)	—	$\hat{\alpha}_c = 2.000/2.127/2.309/2.429$ at $\beta=0/0.5/1.0/1.5$ ($K=3$, $N=250$); 2.160 at $\beta=1, \rho=0.3$
06	band-limited sinusoidal (flat-band K_{eff})		PARTIAL	confounded first pass; corrected sweep slope 0.45 (under-budgeted)	preflight-valid. 2.755/2.911/3.156 at $K_{\text{eff}} \approx 8/16/32$ ($N=500$, slope 0.875); $N=1000$: 2.729/2.971/3.171
07	cross-kernel universality (two-axis)		PARTIAL	v13 alpha row (2.45/2.80/3.20); radial “bound= 4”; see §2	no K_{eff} -only collapse; gabor>gauss by +0.31– 0.57; slope $d\alpha_c/d \ln \ln K_{\text{eff}}$ 0.95 $\rightarrow$ 0.74 $\rightarrow$ 0 with Q
08	Gaussian ker- nel (low-pass baseline)		PARTIAL (growth law)	mixed-fidelity first ladder ($K_{\text{eff}}=4, 8$ at 2.59/2.61)	2.46/2.68/2.91/3.16 at $K_{\text{eff}}=4/8/16/32$ ($N=400$, 16 seeds); mechanism resolved, asymptotic form theory- imposed ($\ln K$ preferred to $K_{\text{eff}} \leq 32$)
09	Gabor kernel (band-pass ra- diality)		CONFIRMED (RS + sign)	omega0= argv bug railed cells to $K_{\text{eff}} \approx 499$ (fixed)	findable ≈ 3.55 ; matched- K_{eff} low $\rightarrow$ band 3.15 $\rightarrow$ 3.25 ($K_{\text{eff}}=32$); radial RS $\alpha_c^{\text{RS}}=4.598$
10	alpha kernel (causal)		PARTIAL	supersedes #7 v13 al- pha row (now v14)	2.00/2.34/2.72 at $K_{\text{eff}}=16.2/32.7/65.7$ ($N=500$); trunc. leaves $\hat{\alpha}_c$ flat
11	rectangular kernel (smoothness boundary)		CONFIRMED— (finite-K benchmark) + PARTIAL (growth law)	—	dt-saturation flat 2.76; benchmark 2.49/2.68/2.75/2.98/3.28 at $K=2/3/4/8/16$ ($N=500$)
12	pure single- frequency sinusoid		PARTIAL	—	flat ≈ 3.4 (3.40/3.40/ $\geq 3.2/3.80$ at $K_{\text{eff}}=8/16/32/64$, $N=500$); RS 4.598; restart 3.40 $\rightarrow$ 3.60

Table 2: Result ledger B — run configuration. “thr.”=threshold rule (median=balanced median of $\max_t V / V_\beta$, $\mathbb{P}(\text{fire}) = \frac{1}{2}$). Density is mean spikes/afferent unless noted. “n/a” = not applicable to a convex/analytic measurement; “n/r” = not recorded in the paper.

#	N	density	seeds	α -grid	learner	epochs	thr.
01	500	n/a (direct Gaus- sian fields, $\rho=0$)	16 ($K2, 4$) / 12 ($K16, 32, 64$)	0.2–0.3 step, per- K bracketed	Adam+cosine lr0.1 b32, 3 restarts, $\beta=16$ softmax	6000 (patience=epochs)	median
02	400 (lad- der 100– 800)	n/a (Gaus- sian pat- terns)	24	≈ 0.06 – 2.2, $\kappa \in$ $\{0, .25, .5, 1, 1.5, 2\}$	convex max- margin QP (quadprog), no learner	n/a	n/a (κ margin)
03	500 (lad- der 250/500/1000)	n/a ($\rho=0$ fields)	16 (8 restarts pre- flight)	fixed $\alpha=2$ (geometry)	reuses #1 hard- max ($\beta=16$)	6000	median
04	400	2.0	16	1.8–2.8 step 0.2	Gütig– Sompolinsky rule, Adam lr0.1 b16 co- sine, 3 restarts	≤ 8000	median
05	250 (lad- der 250/500/1000)	n/a ($K=3$ equicorr. fields)	16	fine (step 0.07 in finite- N)	Adam+cosine smooth-hinge, $n_{\text{restart}}=1-3$	8000– 22000 (slow; $\sim 10^4$)	median
06	500 (lad- der 500/1000/1500)	2.0 (fixed)	n/r (per- (N, K_{eff}) sharded)	per-cell bracketed	Adam+cosine lr0.1 b16, $n_{\text{restart}} \geq 1$	10000/14000/18000	median
07	500	2.0	16 (8+8 sharded)	0.1–0.2, 2.2–4.4 ra- dial end	Adam+cosine b16, $n_{\text{restart}}=2$, lr per kernel (0.1/0.05)	8000	median
08	400	3.0 (fixed)	16	spacing 0.2	Adam+cosine lr0.1 b16, ≥ 3 restarts	3500–5000	median
09	500 ($N=200$ pre- flight)	3.0 (fixed)	n/r	per-cell (2.8/3.1/3.4/4.0)	Adam+cosine lr0.1 b16, ≥ 3 restarts	3500–5000	median
10	500	3.0 (fixed)	n/r	per-cell, grid res. ~ 0.3	Adam+cosine lr0.1 b16, ≥ 3 restarts	n/r (3500– 5000)	median
11	500 (sat. test 300/600)	$r=3$ (fixed)	n/r	step 0.4	Gütig– Sompolinsky minibatch, Adam lr0.1 b16 cosine, ≥ 3 restarts	3500–5000	median
12	500	3.0 (fixed)	16/32 (restart probe)	per-cell	Adam+cosine lr0.1 b16, ≥ 3 restarts	3500–5000	median

Table 3: Run provenance: what was wrong and the fix.

where	what was wrong	fix / current authoritative value
#6 (1st pass)	input density co-varied with K_{eff} (mean spikes = $0.05 K_{\text{eff}}$): at $K_{\text{eff}}=8$ only 0.4 spk/afferent (starved) vs 3.2 at $K_{\text{eff}}=64$; flat $\hat{\alpha}_c \approx 2.1$ read as “no lift”	hold density fixed at 2 spk/afferent; validated learner (critical slowing); add density control. Real monotone lift appears.
#6 (corrected sweep)	slope ≈ 0.45 was an <i>under-budgeting</i> artifact: 59% of solved cells need > 8000 epochs (the cap)	preflight-validated learner, budget 10–18k epochs; slope rises to ≈ 0.875 ($N=500$), 0.913 ($N=1000$); per-cell N -flat.
#9 (Gabor matched- K_{eff})	one-line <code>omega0=</code> argv bug: carrier silently dropped to $\omega_0=1.0$, railing <i>every</i> cell to $K_{\text{eff}} \approx 499$, so K_{eff} was never controlled	argv parsing corrected (carrier honoured, verified in-run); within-Gabor radiality contrast $\approx +0.10$ at $K_{\text{eff}}=32$ ($3.15 \rightarrow 3.25$), band-pass higher at $K_{\text{eff}}=64$.
#10 (alpha)	v13 cross-kernel alpha row was a by-product of the broad #7 sweep	alpha-specific v14 study supersedes it (next row).
#7 (alpha values)	v13 reported alpha $\hat{\alpha}_c = 2.45/2.80/3.20$ at $K_{\text{eff}}=16/32/64$	superseded by #10 v14: 2.00/2.34/2.72 at $K_{\text{eff}} \approx 16/32/64$ (alpha-vs-gauss gap ≈ 0.7 at $K_{\text{eff}} \approx 32$). #7 table now carries v14; v13 kept for provenance.
#7 (radial bound)	“radial/quadratic bound $\alpha_c = 4$ ” (one-quadrature squared-perceptron; Gratsea–Kasper–Lewenstein; Cruciani)	superseded by the two-quadrature RS value $\alpha_c^{\text{RS}} = 4.598$ (4.660 threshold-optimised), derived and quad-verified in #9/#12; 4 retained only as the historical one-quadrature anchor.
#3 (V2)	formerly listed $\alpha_c \approx 1.20/1.99/2.58/3.11$; the third (2.58) spuriously coincided with the discredited α_0 fit	retracted; correct leading values 1.10/1.39/1.60/1.76 at $K=10^2\text{--}10^5$ (well below empirical $\hat{\alpha}_c$; the gap is the open $O(1)$ subleading).
#1 (early ladders)	patience (400, then 2500) below the ~ 4700 -epoch convergence of the $\beta=16$ learner $\Rightarrow P_{\text{solve}}=0$ everywhere; α -grids initially below the crossings	patience=epochs= 6000 (no early cut); α -grids re-bracketed upward; $K=8$ dropped (timeouts / dead hosts).
#8 (1st ladder)	mixed-fidelity: low- K_{eff} cells run coarser, read out artificially flat ($K_{\text{eff}}=4, 8$ pair at 2.59/2.61)	uniform-fidelity ladder (every cell identical config): monotone 2.46/2.68/2.91/3.16. Growth law resolved (§5).

Table 4: Notation used across the series (consistent definitions pulled from the papers).

symbol	definition
K	bin / mode count; in the discrete reduction the number of time bins ($K = T/\tau$ for the boxcar); the abstract independent-look count.
K_{eff}	Rice zero-crossing effective dimension, $K_{\text{eff}} = T\sqrt{\lambda_2/\lambda_0}$, $\lambda_p = \int \omega^p S(\omega) d\omega$; the certified “effective dimension” axis (exact for band-limited kernels, $\Omega T/\sqrt{3}$ for a flat band).
$K_{\text{PR}} / K_{\text{eff}}^{\text{PR}}$	participation-ratio mode count, $K_{\text{eff}}^{\text{PR}} = (T/2\pi)(\int S)^2/\int S^2$; the <i>power</i> -weighted effective dimension that stays finite when λ_2 diverges (#4: $T(\tau+\tau_s)/(\tau^2+3\tau\tau_s+\tau_s^2)$; #11: $1.5T/\tau$).
D_{PR}	power-weighted count $D_{\text{PR}} = (T/\pi)(\lambda_1/\lambda_0)$, a mean- $ \omega $ axis used in #7 (log-divergent for the boxcar).
D_{eff}	participation (power) dimension $D_{\text{eff}} = (\int S)^2/\int S^2$; #10’s proposed robust expressiveness axis (alpha/gauss ratio $2\sqrt{\pi}/5 \approx 0.71$ at matched K_{eff}).
κ_S	spectral kurtosis $\kappa_S = \lambda_4\lambda_0/\lambda_2^2$; the <i>radiality</i> knob ($\kappa_S = 3$ Gaussian/low-pass, $\rightarrow 1$ band-pass; $= \infty$ for alpha/rect).
ϑ / θ	firing threshold; the <i>balanced (median)</i> threshold sets $\mathbb{P}(\text{fire}) = \frac{1}{2}$ (median of $\max_t V$); for the radial classifier $\vartheta = \sqrt{2\ln 2}$ is the Rayleigh-median radius.
Q	Gabor quality factor $Q = \omega_0\sigma$; $Q \rightarrow 0$ low-pass Gaussian, $Q \rightarrow \infty$ single carrier.
α_c	critical storage capacity P/N as $N \rightarrow \infty$ (the existence/replica value); perceptron anchor $\alpha_c = 2$ (Cover/Gardner).
$\hat{\alpha}_c$	<i>findable</i> critical load: the $\frac{1}{2}$ -crossing of $P_{\text{solve}}(\alpha)$ for a validated finder; a lower bound on α_c .
findable	what a strong gradient finder reaches (zero training error on $\geq \frac{1}{2}$ of seeds); $\hat{\alpha}_c \leq \alpha_c^{\text{exist}}$.
existence	whether <i>any</i> weight vector separates all patterns (feasibility); measured directly only in the convex #2 (QP).
RS	replica-symmetric: single-cluster overlap ansatz; gives the leading capacity, exact at convex/linear anchors, an upper estimate where the version space is non-convex.
RSB	replica-symmetry breaking: the shattered/clustered version space (#3); lowers the existence capacity below the RS value.

Table 5: Dependency adjacency (“ $\rightarrow$ ” = depends on / feeds into).

node	depends on / relation
#1 discrete	keystone; $K=1$ anchor = #2 perceptron
#3 EVT	$\leftarrow$ #1 (max-of- K structure)
#4 GP	$\leftarrow$ #3 (lift); supplies $K_{\text{eff}}^{\text{PR}}$
#6 sinusoidal	$\leftarrow$ #3/#4 (kernel specialisation; exact K_{eff})
#8 Gauss	$\leftarrow$ #3 (EVT lift); low-pass baseline
#11 rect	$\leftarrow$ #3/#4 ($\tau_s=0$ saturation limit)
#7 cross-kernel	$\leftarrow$ #6, #8, #10, #11, #12 (matched- K_{eff} inputs)
#9 Gabor	supersedes #7 radial-bound; radial axis
#12 sinusoid	supersedes #7 radial-bound; degenerate radial limit
#10 alpha	feeds #7 (v14 alpha row); causal kernel
#2 spherical	standalone null control (perceptron limit, $K=1$)
#5 soft-max	standalone β -bridge ($\beta \rightarrow 0 = \#2$; $\beta \rightarrow \infty = \#1/\#3$)